Retirement & Inflation — Cheat Sheet

Financial Algebra · Unit 10: Retirement & Inflation

Essential vocabulary

Inflation
A general rise in prices over time, which reduces what each dollar can buy

Consumer Price Index (CPI)
A government measure of the average change in prices paid by consumers for a basket of goods and services

Purchasing power
The amount of goods and services a sum of money can buy

Real value (today's dollars)
A future amount converted into what it would buy at today's prices

Nominal value
An amount in the dollars of the time, not adjusted for inflation

401(k)
An employer-sponsored retirement savings plan funded by payroll contributions, often with an employer match

Employer match
Money an employer contributes based on the employee's own contributions, up to a limit

Vesting
The schedule by which an employee gains full ownership of employer contributions

Traditional (pre-tax) account
A retirement account funded with pre-tax dollars; withdrawals in retirement are taxed

Roth account
A retirement account funded with after-tax dollars; qualified withdrawals in retirement are tax-free

IRA
An Individual Retirement Account opened by a person rather than through an employer

Nest egg
The total savings accumulated for retirement

4% guideline

Key formulas

Future price	price × (1 + i)^t
Today's dollars	$\frac{\text{future amount}}{(1 + i)^t}$
Inflation from CPI	$\frac{CPI_2 - CPI_1}{CPI_1}$
Real return	$\frac{1 + \text{nominal}}{1 + \text{inflation}} - 1$
Employer match	match rate × min(your %, limit) × salary
Nest egg	$\frac{\text{spending} - \text{other income}}{0.04}$
Balance projection	$FV = PMT \cdot \frac{(1 + i)^N - 1}{i}$

Account types

Traditional 401(k)/IRA	Tax break now; taxed at withdrawal
Roth 401(k)/IRA	Taxed now; qualified withdrawals tax-free
Employer match	Part of pay; may vest over time
Social Security	Partial income replacement

A planning rule of thumb that withdrawing about 4% of savings in the first year of retirement (adjusted for inflation after) may last about 30 years

Social Security
A federal program funded by payroll taxes that provides retirement, disability, and survivor benefits

Real rate of return
A return adjusted for inflation: $\frac{(1 + \text{nominal})}{(1}$ + inflation) − 1

Common misconceptions

- Small inflation does not matter.
- Cash is safest for decades.
- Social Security covers everything.
- Future-dollar projections show real wealth.

Worked example

1. Age 25, $50,000 salary, 6% + 50% match → $375/month
2. At a hypothetical 6% for 42 years ≈ $851,000 (future $)
3. At 2.5% inflation ≈ $302,000 in today's dollars
4. 4% guideline → ≈ $12,000/yr (today's $) + Social Security

Questions to ask before deciding

- Am I getting the full match?
- What is this projection in today's dollars?
- Roth or traditional — is my tax rate higher now or later?
- When does the match vest?
- What return and inflation are assumed?

Decision tips

- Start early and automate.
- Increase contributions 1% per year.
- Always capture the full match.
- Check projections in today's dollars.

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